

Project Initialization and Planning Phase

Date	28 June 2026
Team ID	SWTID-2026-3658
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to develop an AI-powered educational assistant that enhances learning through intelligent question answering, concept explanation, quiz generation, text summarization, and personalized learning recommendations. The system leverages Google Gemini 1.5 Flash and LaMini-Flan-T5 models to deliver accurate, personalized, and interactive educational support through a secure and user-friendly web application.

Project Overview	
Objective	The primary objective is to provide students and self-learners with an AI-powered educational platform that offers intelligent question answering, simplified concept explanations, quiz generation, text summarization, and personalized learning paths.
Scope	The project enables users to register securely, interact with AI models through a responsive web interface, receive educational assistance, generate quizzes and summaries, save learning history, bookmark important responses, and access personalized learning recommendations.
Problem Statement	
Description	Students often struggle to understand complex concepts, find reliable answers, generate practice quizzes, summarize lengthy study material, and organize their learning effectively. Existing educational resources are scattered across multiple platforms, making learning time-consuming and less efficient.
Impact	Solving this problem improves learning efficiency, provides personalized educational support, increases student engagement, and helps learners prepare more effectively through AI-assisted guidance.
Proposed Solution	
Approach	Develop a modular web application using FastAPI , Google Gemini 1.5 Flash , LaMini-Flan-T5 , SQLite , SQLAlchemy , Jinja2 , HTML , CSS , and JavaScript to provide intelligent educational assistance with secure authentication and persistent learning history.

Key Features	- AI-powered Question Answering - Concept Explanation using LaMini-Flan-T5 - Automatic Quiz Generation - Text Summarization - Personalized Learning Path Recommendation - Secure User Registration and Login (JWT + bcrypt) - Learning History and Response Bookmarks - Interactive Responsive Web Interface - REST API Documentation using Swagger UI
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5 Processor
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	256 GB SSD
Software		
Frameworks	Frameworks	FastAPI, Jinja2
Libraries	Libraries	Google Generative AI, Transformers, Torch, SQLAlchemy, bcrypt, python-dotenv, Uvicorn
Development Environment	IDE	Visual Studio Code
Data		
Data	Source, size, format	User educational queries, AI-generated responses, quizzes, summaries, learning paths, SQLite database